



DATA ANALYSIS PROJECT (PYTHON)



Online Delivery Dataset Analysis

◇ 1. Problem Statement

Students must define the objective of the project.

Example: - Which company provides the fastest delivery? - Does delivery time affect customer ratings? - Which city has the highest order value? - Do discounts improve customer satisfaction?

◇ 2. Dataset Description

Explain each column clearly:

- Order_ID: Unique order number
 - Company: Delivery platform name
 - City: Location of the order
 - Customer_Age: Age of customer
 - Order_Value: Total order amount
 - Delivery_Time_Min: Delivery time in minutes
 - Distance_Km: Distance covered
 - Items_Count: Number of items in order
 - Product_Category: Category of product
 - Payment_Method: Mode of payment
 - Customer_Rating: Rating given by customer
 - Discount_Applied: 1 = Yes, 0 = No
 - Delivery_Partner_Rating: Rating of delivery partner
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◇ 3. Data Loading

Students should load the dataset using Python.

◇ 4. Data Cleaning

Students must perform the following steps:

- Identify missing values
 - Handle null values
 - Fix incorrect data types
 - Remove duplicates
 - Handle inconsistent values (e.g., spelling mistakes in city names)
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◇ 5. Exploratory Data Analysis (EDA)

Perform basic exploration:

- Understand dataset structure
 - Analyze distributions of numerical columns
 - Check unique values in categorical columns
 - Identify patterns and anomalies
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◇ 6. Analysis Questions

🌀 Basic (Compulsory)

1. Find average order value per company
2. Count total orders per city
3. Identify most used payment method
4. Calculate average delivery time

🌀 Intermediate

5. Does discount affect customer rating?
6. Which product category has highest orders?
7. Analyze relation between distance and delivery time
8. Which company has best customer ratings?

🌀 Advanced

9. Find top 5 highest value orders
10. Which age group spends the most?
11. Find correlation between:
 - Delivery time and customer rating
 - Distance and delivery time

◇ 7. Data Visualization

Students must create the following graphs:

- Bar Chart → Orders per company
 - Pie Chart → Payment methods
 - Line Chart → Distance vs Delivery time
 - Heatmap → Correlation matrix
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◇ 8. Key Insights

Students must write at least 5 insights based on analysis.

Example: - Faster delivery leads to better ratings - Some companies perform better in specific cities - Discounts do not always improve ratings

◇ 9. Conclusion

Summarize findings: - What did you learn? - Which company performs best? - Any recommendations?

Final Submission Requirements

Students must submit:

1. Python Notebook (.ipynb)
 2. Cleaned dataset (CSV)
 3. Presentation (5–7 slides):
 - Problem Statement
 - Data Cleaning
 - Analysis
 - Visualizations
 - Insights
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Important Instructions

- Focus on insights, not just code
- Use proper graphs and explanations
- Keep work clean and well-structured
- Explain each step clearly

End of Assignment